

Deepak Mallampati

FORWARD DEPLOYED ENGINEER / APPLIED AI

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Kent, OH (open to NYC relocation) -- F-1 OPT, authorized to work now

PROFESSIONAL SUMMARY

Applied AI engineer (4+ years) who ships production AI agents and RAG systems in regulated healthcare end-to-end — from customer scoping through production support. At Progress Solutions, built agentic LLM workflows on a conductor-subagent orchestration framework that cut token consumption **42%** while sustaining task accuracy; at Vanguard, shipped **25+ AI agents** and **12 APIs** to **20+ internal teams** serving **100K+ requests/day** at **800ms p95**. Healthcare data discipline: privacy-preserving clinical pipelines (k-anonymity, l-diversity, Laplace DP) cut re-identification risk from **3.38% to 0.32%** while retaining **99% baseline predictive performance**. Comfortable embedding with non-technical operators, translating EHR/voice/scheduling friction into deployed agents.

CORE COMPETENCIES

Forward-Deployed Customer Delivery . Healthcare AI Agents (Voice + Text) . Agentic LLM Systems (Conductor-Subagent) . Production RAG & LLM Evaluation . Python / FastAPI / Microservices . EHR-Adjacent Privacy & Compliance . Third-Party API Integration . Multi-Stakeholder Engineering Delivery . End-to-End Production Ownership

EXPERIENCE

Progress Solutions

Jul 2025 -- Present

AI Engineer -- USA | Healthcare AI

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex healthcare objectives into context-scoped subagents that execute autonomously -- reducing token consumption by **42%** while sustaining task accuracy across multi-step clinical workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Designed and operationalized **privacy-preserving data workflows** for sensitive clinical datasets, implementing K-anonymity, L-diversity, and differential privacy to enable compliant analytics and model development with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%** and stabilizing live customer deployments.
- Established a **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to surface regressions before release.

Vanguard

Jan 2024 -- Jan 2025

AI Engineer Intern -- USA | Regulated FinTech

- Developed and shipped **25+ AI agents** and workflows for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across multiple regulated business units.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** across the same task suite, improving task success rates by **27%** on internal evaluation datasets and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities -- reduced p95 latency from 1.5s to **800ms** and accelerated feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across **20+ internal teams**.

Kent State University

Oct 2023 -- Jan 2024

ML & Gen AI Research Assistant -- Ohio, USA | Healthcare Research

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset, applying k-anonymity (k=3), l-diversity (l=2), and Laplace differential privacy across four epsilon levels (0.1-2.0).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure (0.94% -> 0%), while retaining **99% of baseline predictive performance** (RF AUC 0.689 vs. 0.696; weighted-F1 0.895).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated 361-example instruction dataset, conditioning on a six-stage narrative schema for controllable long-form generation.
- Authored a peer-review-ready **IEEE-format conference paper** documenting methodology and results.

Emerson

Jun 2022 -- Apr 2023

ML Trainee -- India | Industrial ML

- Developed and trained supervised regression and classification models on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction on domain-specific financial datasets, achieving **89% F1-score** on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**. Demonstrates fast solo ship from problem to production, multi-provider AI vision integration, and customer-facing third-party API delivery.

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision. Conductor-subagent orchestration with structured outputs and eval framework -- direct analog to forward-deployed customer-automation work.

Privacy-Preserving Clinical ML Pipeline

Framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy. Quantifies privacy-utility trade-off explicitly -- direct EHR-adjacent proof for healthcare AI deployments.

EDUCATION

M.S. in Computer Science -- Kent State University, Ohio

2025 | GPA 3.70

TECHNICAL SKILLS

Languages & Frameworks: Python (primary), TypeScript, Java, SQL | FastAPI, React | PyTorch, TensorFlow

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG, agentic workflows, prompt engineering, content safety, LoRA/QLoRA fine-tuning

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Cross-encoder re-ranking | Semantic + hybrid search | Document chunking strategies

MLOps & Cloud: AWS (Bedrock, SageMaker, S3, Lambda), Azure (OpenAI, AI Services), GCP Vertex AI | Docker, Kubernetes | CI/CD | MLflow, W&B

Integration & APIs: REST/microservices, FastAPI, third-party API integration, observability/logging, structured outputs, schema-validated agent contracts

Regulated Domains: HIPAA-conscious governance, PII redaction, audit logging, privacy-preserving ML (k-anonymity, l-diversity, differential privacy) -- direct EHR/FHIR/HL7-adjacent foundation

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required